

Atanu Kundu

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Research Area

- 📌 I am a researcher in formal verification and falsification of cyber-physical systems, with expertise in developing algorithms for bounded model checking (BMC) and machine learning-based falsification. Key contributions include:
- Developed a CEGAR-based BMC algorithm for affine hybrid systems, which integrates SAT-solving based path enumeration and path-wise reachability analysis, enabling efficient verification of safety properties.
 - Designed a bounded reachability algorithm for compositional affine hybrid systems, specifically tailored for compositional systems, thereby improving scalability in the analysis of large-scale CPS.
 - Designed surrogate model-based falsification frameworks for industrial CPS, bridging data-driven techniques with robustness monitoring to address the limitations of traditional optimization-based falsification methods. In particular, deep neural networks and decision trees are employed as surrogate models of the system under test.
 - Developed a causation-aware falsification strategy leveraging reinforcement learning and causation monitoring to efficiently accelerate counterexample searching in CPS.

Education

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| 2020 – present
<i>thesis submitted</i> | 📌 PhD in Computer Science from IACS Kolkata, India.
Thesis title: <i>Bounded Model Checking and Data-Driven Falsification strategies for CPS.</i>
Courses: Big Data Analytics and Advanced Machine Learning CGPA: 8.50 |
| 2016 – 2018 | 📌 M.Sc. in Computer Science from Visva-Bharati, India.
Courses: Advanced Data Structure, AI, DAA, etc CGPA: 7.93 |
| 2013 – 2016 | 📌 B.Sc. in Computer Science from The University of Burdwan.
Percentage: 66.12 |

Research Publications

- 1 S. Gon, A. Kundu, and R. Ray, “Falsification of cyber-physical systems using causation-aware reinforcement learning,” *accepted in 29th ACM International Conference on Hybrid Systems: Computation and Control, HSCC/ICCPS*, 2026.
- 2 L. Bu, A. Kundu, R. Ray, and Y. Shi, “Arch-comp25 category report: Hybrid systems with piecewise constant dynamics and bounded model checking,” in *Proceedings of 12th Int. Workshop on Applied Verification for Continuous and Hybrid Systems*, G. Frehse and M. Althoff, Eds., ser. EPiC Series in Computing, vol. 108, EasyChair, 2025, pp. 1–14. 📄 DOI: 10.29007/x3nv.
- 3 T. Khandait et al., “Arch-comp25 category report: Falsification,” in *Proceedings of 12th Int. Workshop on Applied Verification for Continuous and Hybrid Systems*, G. Frehse and M. Althoff, Eds., ser. EPiC Series in Computing, vol. 108, EasyChair, 2025, pp. 169–189. 📄 DOI: 10.29007/dgnn.

- 4 A. Kundu, S. Gon, and R. Ray, "Data-driven falsification of cyber-physical systems," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2025.  DOI: 10.1109/TCAD.2025.3608632.
- 5 L. Bu, A. Kundu, R. Ray, and Y. Shi, "Arch-comp24 category report: Hybrid systems with piecewise constant dynamics and bounded model checking," in *Proceedings of the 11th Int. Workshop on Applied Verification for Continuous and Hybrid Systems*, G. Frehse and M. Althoff, Eds., ser. EPiC Series in Computing, vol. 103, EasyChair, 2024, pp. 1–14.  DOI: 10.29007/nv67.
- 6 T. Khandait et al., "Arch-comp 2024 category report: Falsification," in *Proceedings of the 11th Int. Workshop on Applied Verification for Continuous and Hybrid Systems*, G. Frehse and M. Althoff, Eds., ser. EPiC Series in Computing, vol. 103, EasyChair, 2024, pp. 122–144.  DOI: 10.29007/hgfv.
- 7 A. Kundu, S. Gon, and R. Ray, "Data-driven falsification of cyber-physical systems," in *Proceedings of the 17th Innovations in Software Engineering Conference*, ser. ISEC '24, , Bangalore, India, Association for Computing Machinery, 2024, ISBN: 9798400717673.  DOI: 10.1145/3641399.3641401.
- 8 A. Kundu, S. Das, and R. Ray, "Sat-reach: A bounded model checker for affine hybrid systems," *ACM Trans. Embed. Comput. Syst.*, vol. 22, no. 2, Jan. 2023, ISSN: 1539-9087.  DOI: 10.1145/3567425.
- 9 C. Menghi et al., "Arch-comp23 category report: Falsification," in *Proceedings of 10th International Workshop on Applied Verification of Continuous and Hybrid Systems (ARCH23)*, G. Frehse and M. Althoff, Eds., ser. EPiC Series in Computing, vol. 96, EasyChair, 2023, pp. 151–169.  DOI: 10.29007/6nqs.
- 10 L. Bu, G. Frehse, A. Kundu, R. Ray, Y. Shi, and E. Zaffanella, "Arch-comp22 category report: Hybrid systems with piecewise constant dynamics and bounded model checking," in *Proceedings of 9th International Workshop on Applied Verification of Continuous and Hybrid Systems (ARCH22)*, vol. 90, 2022, pp. 44–57.  DOI: 10.29007/lnzf.

Under Review

- A. Kundu, P. Sarkar, R. Ray, "A CEGAR-centric Bounded Reachability Analysis for Compositional Affine Hybrid Systems ".

Projects and Work Experiences

- 2017 - May 2018 ■ Ray TCP over wireless network with free-space optical and worldwide interoperability for microwave access.
under the supervision of Mr. Subhasis Banerjee, Visva-Bharati, India.
- May - Dec 2022 ■ Building neural network models of the standard Cyber-Physical Systems (CPS) benchmarks.
Under the supervision of Dr. Rajarshi Ray, IACS Kolkata, India.
Building Feed-forward Neural Networks (FNNs) for CPS benchmarks such as hybrid automata and Simulink models. We built FNN models for systems with both linear and non-linear dynamics, as well as time-varying input signals. These models are approximations of the original models and can be useful in developing applications for such systems.

Skills

- Languages ■ Native language Bengali. Fluent in English and Hindi.
- Coding ■ Proficient in C, C++, Python, and SMT-LIB2. Familiar with MATLAB and LaTeX.
- Technical Domains ■ Formal Verification, Hybrid Automata, Falsification, Reachability Analysis, SAT Solving, STL, Robustness Monitoring, and Surrogate Modeling using Machine Learning.

Skills (continued)

DevOps	■ Experience in git and Docker.
Tools	■ FlexiFal, NNFal, XSpeed, SAT-Reach, SpaceEx, Flow*, dReach, and breach.
Soft Skill	■ Team Work, Time Management, Presentation Skills, and Leadership.
Misc.	■ Academic research, teaching, training, consultation, LaTeX typesetting, and publishing.

Teaching Experiences

Spring 2022	■ Teaching assistant on Artificial Intelligence (COM 4211) Responsibilities included setting and evaluating lab assignments. Supervising students throughout the course to succeed in the lab assignments.
Autumn 2023	■ Teaching assistance on Object-Oriented Programming with C++ Lab (COM 4111) My duties involved creating and assessing assignments and surprise tests. Additionally, I was responsible for overseeing students' progress during the course, ensuring their success in laboratory assignments.

Miscellaneous Experience

Selected Talks

2025	■ Research paper Data-driven Falsification of Cyber-physical Systems accepted at IEEE TCAD 2025 and presented at RHPL co-located with FSTTCS 2025, BITS Pilani, GOA.
2024	■ Research paper Data-driven Falsification of Cyber-physical Systems accepted at ISEC 2024 and presented the paper at IIIT Bangalore.
2023	■ A short presentation titled A Framework for Detecting Unsafe Behavior in Cyber-Physical Systems presented in Tech Symposium on Computing Trends 2023, organized by THALES associated with UNIVERSITY OF CALCUTTA.
2022	■ SAT-Reach: A Bounded Model Checker for Affine Hybrid Systems presented in the formal methods update meeting 2022 at IIT Delhi.

Certification & Awards

2026	■ Awarded ACM SIGBED travel grant for attending CPS-IoT Week 2026 .
2023	■ Special recognition for completing a short thesis presentation in exactly 5 minutes. Awarded by THALES associated with UNIVERSITY OF CALCUTTA.

References

Dr. Rajarshi Ray

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